

Beyond the Box Score: Discovering What Really Wins Baseball Games

Alisa Crowe, Rita Herfi, and Samantha Keppler

Big Data Analytics, San Diego State University

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Ryan Lafler

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Abstract

Baseball is a sport defined by numbers—batting averages, earned run averages, on-base percentages, and countless other statistics that attempt to measure every aspect of the game. However, not all of these numbers carry equal weight when it comes to building a winning team. This project investigates which statistics most directly contribute to team success by analyzing a variety of player and game-level metrics. Using data analysis and visualization techniques, the study explores how offensive, defensive, and pitching statistics interact to predict team wins. It also explores fan sentiment about teams and whether or not it reflects real-life performance outcomes. With this information, a chat bot will be created that communicates insights to a user.

***Keywords:* Baseball, Sentiment Analysis, Text Mining, Predictive Modeling, Chatbot**

1. Introduction

Baseball is a game rich with numbers. Every swing, pitch, and play generates new statistics, from batting averages and on-base percentages to strikeout rates and fielding metrics. Yet for all this data, one central question remains: how do these statistics come together to create a winning baseball team? Beloved by millions of fans, baseball can bring intense excitement with every win and deep disappointment with every loss. With so many stats recorded every day, it can be difficult to determine what truly defines a “good” team, especially when individual games and seasons are so unpredictable.

This project aims to make sense of that complexity by analyzing baseball data to simplify and clarify player and team performance for fans and analysts. Rather than manually tracking every statistic, data is used to model and predict possible outcomes, providing more accurate insights into future game results and overall performance. The analysis draws on both unstructured text and structured numerical data, collected through web scraping, and a Kaggle baseball data bank. All data processing, analysis, querying, and prediction are conducted using Python and its libraries, forming a comprehensive approach to understanding how

statistics can help explain, and even forecast, success in baseball.

2. Data Sources

The two primary sources of data used for this project are structured numerical data and unstructured textual data. The numerical data was obtained from the *Baseball Databank* dataset on Kaggle, which contains historical and contemporary statistics for all MLB players and teams from 1871 to 2015. The main file used from this dataset, *Master.csv*, includes records for 18,846 players with 24 attributes such as player IDs, birth and death dates, full names, nicknames, and other biographical information. Across all fields, the dataset contains 63,324 missing values, reflecting the long historical span and variability in early baseball record-keeping. To support the collection of textual sentiment data, an additional column in *Master.csv* was generated that enumerated possible name variants, such as nickname + last name or first initial + last name, to better capture mentions of players on social media.

The unstructured textual data consists of social media posts scraped from Reddit and

BlueSky using their respective APIs. Queries were generated using (i) team names, including common team abbreviations and nicknames and (ii) player names and name variants extracted from *Master.csv*. This process yielded one month of fan discourse at both the team and player levels. After collection, the data was stored in CSV format, resulting in 30,217 player-level posts and 119,648 team-level posts. Each row contains a post ID, platform, subreddit (for Reddit posts), author username, raw text, hyperlink, timestamp, matched keyword, and character length. Because the textual data are user-generated, they include informal language, slang, sarcasm, domain-specific terminology, and noise typical of sports conversations. These characteristics required substantial preprocessing including tokenization, lemmatization, and stop-word removal, prior to downstream sentiment and emotion modeling.

3. Methods

3.1 Data Preprocessing

To provide the basis for team-level analysis, the Reddit team dataset and the Bluesky team dataset were combined into a single team-level dataset. Then, each record was

assigned to a team based on comparing the text field to a curated list of mapped keywords that corresponded to a full, standardized team name. For example, “Yankees”, “NYY” and “Yanks” were all mapped to “New York Yankees” by applying exact and fuzzy matching through pandas string methods. Then, fields were added for 2025 team-level statistics (such as win percentage, run differential, and batting average) using a python dictionary that mapped statistics to each standardized team name.

The next step was to add sentiment and emotion-related columns to the dataset. This was only done to the team-level data, excluding the player-level data, because there would be stronger and more interpretable signals there.

3.2 Sentiment Scoring

Sentiment polarity was calculated using the VADER (Valence Aware Dictionary and Sentiment Reasoner) model from the NLTK library. This was chosen because it is specifically designed for analyzing social-media text, making it well-suited for Reddit-style posts.

For each record, the model outputted a continuous sentiment score in the range of -1 to 1, with -1 representing the most negative

sentiment and 1 representing the most positive sentiment.

3.3 Emotion Classification

To analyze the emotional tone of each post, a simple keyword-matching approach was utilized based on an emotion dictionary called the NRC Emotion Lexicon (EmoLex), which contains lists of words associated with emotions such as anger, joy, fear, and sadness. This model works by breaking down each post into individual words and comparing these tokens against the emotion lists. Emotion labels were added if a post had more words corresponding to one emotion than any other emotion.

Following emotion labeling, team-level aggregates were calculated in order to explore emotional tone in fanbases as a whole rather than individual posts.

3.4 Theme Extraction and Topic Classification

A keyword-based topic tagging system was used to assign a discussion topic to each post. The theme dictionary was created based on keyword frequency analysis using pandas and collections, counter, manual inspection of common n-grams, and team-specific terms. Each

post was assigned to one of the following themes: team performance, player performance, fan experience, humor & memes, management decisions, league fairness & policies, offseason & outlook, media & analysis, off-topic/cross-sport, and baseball culture and nostalgia.

Because it is certainly possible that an individual post hits upon multiple of these themes, each post was only labeled with the most relevant themes according to rule-based matching implemented with pandas and re. For each team, the share of posts in each theme was computed to be used later to compare discourse and thematic valence across fanbases.

3.5 Sentiment Volatility Analysis

Sentiment volatility was calculated to understand how consistent sentiment is across each fanbase. It was computed using pandas as each team's standard deviation of post-level sentiment:

$$[\sigma_{\text{sentiment}} = \text{std}(\text{compound sentiment scores})]$$

3.6 Exploratory Data Analysis (EDA)

EDA began with various checks on variable types, missing values, presence of all 30

MLB teams in the dataset, and removal of outlier or bot-related posts.

Next, the distribution of all variables in the dataset were investigated, helping to ensure that real-life baseball variables were correlated as expected. This also helped verify that the generated emotion-related columns held real and interpretable signals that could be examined later.

Next, statistical testing was conducted on the aggregated team-level fields to check whether there were systematic differences in fan sentiment, emotion, and discussion themes across teams. For each variable, one-way analyses of variance (ANOVAs) assessed a null hypothesis that all teams showed equal means against the alternative hypothesis that at least one team differed. When significant effects were found, Tukey's Honest Significant Difference (HSD) was used to determine which team pairs were significantly different from others. This analysis helped quantify variability between teams and set the stage for selection of variables in the modeling process of the study.

3.7 Team Success Modeling

Since the dataset contained fan discourse rather than real-life observations, all added statistics were aggregated to the team-year level, ensuring that each row in the dataset was assigned to a single MLB team. The features used in both models included all of the team-level statistical columns.

All predictors were standardized prior to modeling to ensure that effects were on a comparable scale. Additionally, both models were trained and evaluated using a common 80/20 train/test split.

For the Win Percentage Model, a multiple linear regression model was used to predict how each metric individually affected the assigned team's overall win percentage. This model was deemed appropriate because the target variable is continuous and the relationships between the target variable and the other variables were assumed to be linear. Then, feature importance was calculated using standardized regression coefficients.

For the Playoff Qualification Model, a logistic regression classifier was fitted to estimate the probability that a team reached the playoffs based only on its aggregated

performance metrics. This kind of model was chosen because the target variable was binary and the relationships between the target variable and the other variables were nonlinear. Then, feature importance was evaluated using standardized log-odds coefficients.

3.8 Tools

Exploratory visualization was done in JupyterHub using matplotlib and other similar visualization packages. This environment was also used for data cleaning, sentiment scoring, emotion and theme labeling, statistical modeling, and various other preprocessing steps. Final visualizations were created using Tableau.

3.9 Interactive Chatbot

To make the analytical insights from this study accessible to users, an interactive chatbot was developed that leverages retrieval-augmented generation (RAG) to answer questions about MLB teams, players, statistics, and fan sentiment. The chatbot system consists of a FastAPI backend that handles natural language queries, a React-based frontend that provides the user interface, and a knowledge base for the model to retrieve from.

The knowledge base for the chatbot was constructed by aggregating and processing the analytical outputs from the study. Team-level summaries were created by grouping social media posts by team and season, then computing aggregated performance statistics, sentiment metrics, emotion distributions, and theme profiles for each team-year combination. Similarly, player-level summaries were generated by aggregating posts mentioning individual players along with their corresponding performance statistics. These summaries, along with global insights documents, example fan posts, a glossary of baseball metrics, frequently asked questions, and methodology documentation, were organized into seven distinct document collections to be retrieved from the chatbot.

Document indexing was performed using ChromaDB, which is a vector database designed for semantic search. Each document was split into appropriately-sized chunks, with markdown files segmented by heading structure and JSON files processed as individual records. The nomic-embed-text embedding model from Ollama was used to generate vector

representations of each document chunk. These embeddings capture semantic meaning, allowing the system to retrieve relevant information even when user queries use different terminology than the source documents.

Query processing uses an intelligent routing system that analyzes incoming questions to determine which document collections are most relevant using cosine similarity. The routing logic identifies mentions of specific teams or players using keyword matching against a comprehensive mapping of team names, abbreviations, and common nicknames. When a team or player is detected, the system prioritizes searching the corresponding team or player collections and applies the corresponding metadata filters to ensure precise retrieval. The system also recognizes queries about sentiment, emotions, or themes and routes them to collections containing global insights and analysis documents. For questions about metric definitions or methodology, queries are directed to the glossary and edge cases collections.

Once relevant documents are retrieved through semantic similarity search, the top results are formatted into a context string that

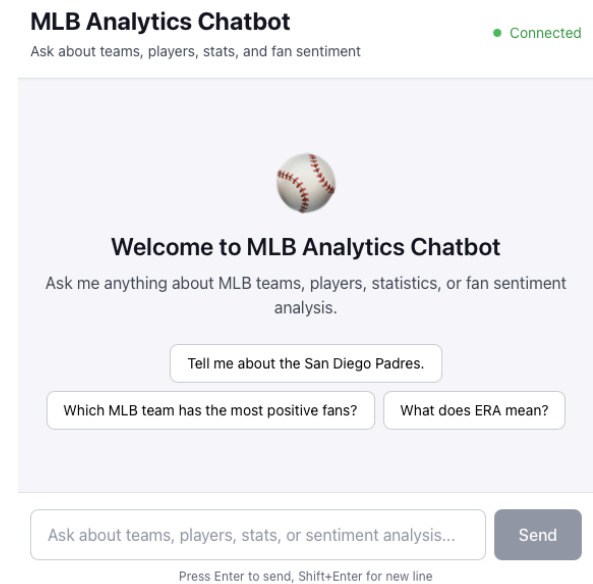
includes source metadata such as team names, player names, and document sections. This context, along with the user's query and recent conversation history, is passed to a large language model (LLM) running on Ollama. The llama3:8b model was used for text generation, which was chosen for its balance between response quality and computational efficiency. The LLM is provided with a system prompt that instructs it to answer questions using only information from the retrieved context, distinguish between objective statistics and subjective sentiment, and acknowledge when information is not available in the knowledge base.

The backend API exposes two main endpoints: a health check endpoint that verifies connectivity to both Ollama and ChromaDB, and a chat endpoint that accepts user messages and conversation history, performs retrieval, generates responses, and returns both the answer and source citations. The frontend interface, built with React and TypeScript, provides a conversational chat interface where users can ask questions, view responses with source attribution, and see system status indicators.

This RAG-based approach ensures that the chatbot's responses are grounded in the actual data and analysis from the study, rather than relying solely on the LLM's training data or fine-tuning, which is far more resource intensive (Soudani et al., 2024). By combining semantic search with generative language modeling, the system can provide contextually relevant answers about MLB analytics while maintaining transparency about its knowledge sources.

Figure 1

User Interface for MLB Analytics Chatbot



4. Results

4.1 Team Success Modeling

To explore what factors affect on-field outcomes, two predictive models were

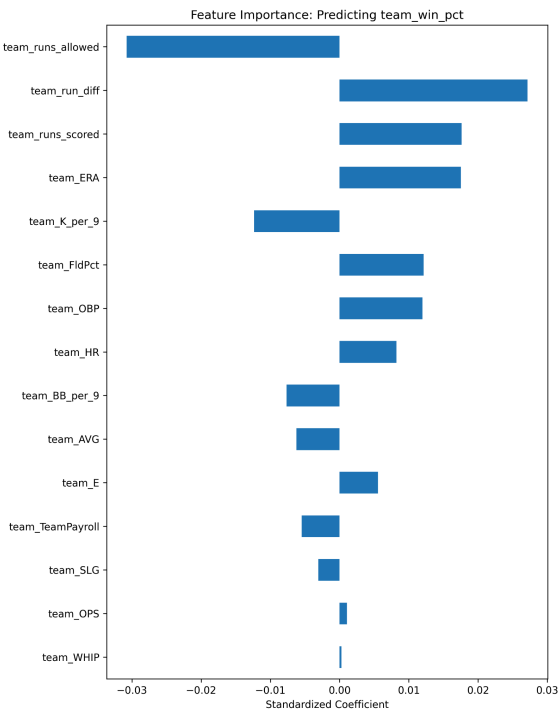
developed using only team-level performance statistics from the 2025 MLB regular season. Because both models displayed high accuracy in performance evaluation, feature engineering or parameter tuning were not needed.

4.1.1 Predicting Win Percentage (Regression Analysis)

With the help of standardized performance metrics, a multiple linear regression model was fit to predict win percentages of teams.

Figure 2

Feature Importance for Predicting Win Percentage



The model achieved an R^2 of ~ 0.9 and an RMSE of ~ 0.014 , showing a very high rate of predictive power.

A surprising effect found by this model was the negative coefficient associated with strikeouts per nine innings (K/9), a common pitching metric. Higher K/9 was expected to be associated with better pitching performance and therefore more successful outcomes for the team as a whole. However, within the context of this multivariate regression model, this discrepancy can likely be explained by the interaction between features. For one, positive predictive power may be getting absorbed by other pitching metrics such as ERA and WHIP that are also present in this dataset. Another possible explanation would be that K/9 cannot explain pitching quality in a vacuum. Sometimes, teams with higher strikeout counts can also show higher rates of walks, increased pitch counts, or heavy reliance on bullpen relievers. Therefore, K/9 may be more reflective of the strengths of individual pitchers, and not the efficacy of the team's pitching fleet as a whole.

Across both models, defensive and pitching metrics played a larger role in

determining team success than did offensive metrics. For both, the strongest predictive variable was runs allowed. While these results are consistent with sabermetric research (Tango et al., 2007), it may be counterintuitive to the average baseball fan, therefore making this a very useful insight for building the LLM.

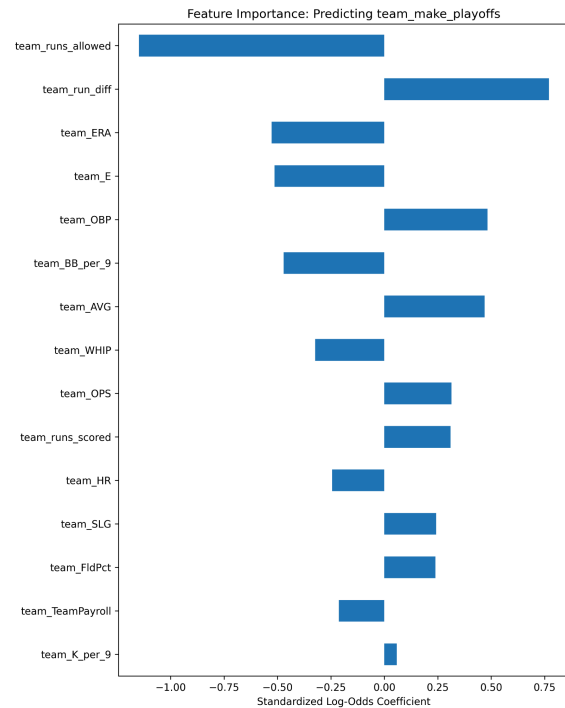
4.1.2 Predicting Playoff Qualification (Classification Analysis)

Next, using a logistic regression classifier, modeled the probability of a team making the playoffs. This model achieved even higher rates of accuracy, with a ROC AUC of ~0.97 and an overall accuracy of 92% on the training data, showing that clear distinctions could be made between teams that made the playoffs and teams that did not.

Similar to the win-percentage model, the most important feature here was runs allowed, once again highlighting the importance of preventing runs in overall team success. Other strong predictors include run differential, ERA, fielding percentage, and WHIP. Offensive metrics, such as OBP, OPS, and total runs scores displayed small but significant coefficients on the team's competitive outcomes.

Figure 3

Feature Importance for Predicting Playoff Qualification



Surprisingly, payroll showed a slightly negative effect, indicating that teams with high budgets may be underperforming recently. The figure below shows the feature importance for this model.

4.2 Fan Sentiment Analysis

In an attempt to discover how fan attitudes are correlated with team performance, sentiment measures were analyzed, discrete emotion labels, discussion themes, and topic clusters of each online post in the dataset. The

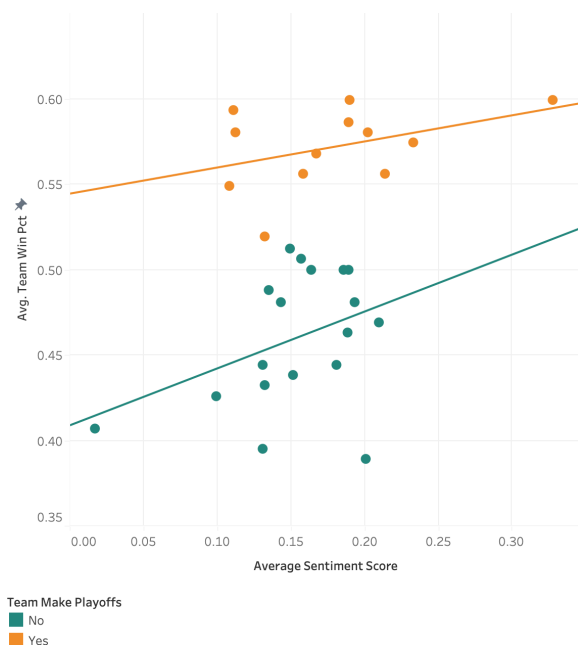
results showed that fan sentiment does indeed reflect team performance in terms of winning and losing, but emotion and theme analysis provide a significantly richer signal.

4.2.2 Sentiment and Team Outcomes

For each team, average sentiment scores fell within a relatively narrow band, but displayed a consistent correlation with team success metrics. As shown in the below figure, teams that reached the playoffs showed both higher sentiment scores and higher win percentage while teams that did not reach the playoffs had significantly more negative sentiment.

Figure 4

Sentiment vs. Team Wins



A similar pattern occurred when sentiment was compared with different team success metrics, such as run differential. Run differential is calculated by, "subtracting the total number of runs (both earned and unearned) the team has allowed from the number of runs it has scored." (Major League Baseball, n.d.). Although the magnitude of the correlations is moderate, the trend is consistently indicative that higher fanbase sentiment reflects higher performance of the corresponding team.

And similarly, teams displaying negative performance metrics, such as the Athletics, Rockies, or Nationals, had unsurprisingly negative sentiment from their fanbases.

To support these trends, regression analyses were run that indicated sentiment exhibits small but reliably positive coefficients for wins, win percentage, and run differential. Therefore, while sentiment should not be viewed as a standalone predictor of performance, it shows a directional correlation with team performance.

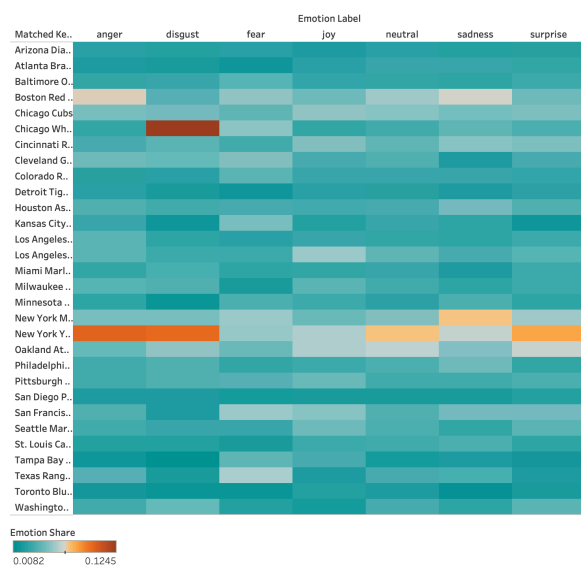
4.2.2 Emotion Signals in Fan Discourse

The discrete emotional analysis provided more color than the scalar sentiment

score. Figure 4 exhibits how frequently each emotion appeared in posts about each team. Although the majority of posts were classified as “neutral” due to the high frequency of sports-reporting on social media, several teams showed noticeably higher rates of negative emotions such as anger, disgust and sadness.

Figure 5

Emotions by Team



Further analysis of each emotion revealed systematic correlations with performance outcomes:

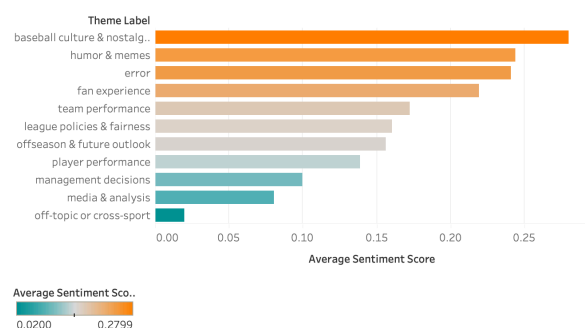
- **Joy** was disproportionately reflected in posts about teams that made the playoffs and showed strongly positive run differentials.

- **Disgust** and **Fear** appeared the most with underperforming teams with losing records.
- **Neutral** tone was often associated with analytical discussion but was also weakly correlated with below-average performance of teams.
- **Anger** was associated with both weak and strong performance of teams, particularly in fanbases with higher expectations, indicating frustration with singular game outcomes or managerial decisions rather than actual team weakness.

Overall, these patterns suggest that emotional tone, rather than general valence, better differentiates teams with successful seasons from those with unsuccessful seasons.

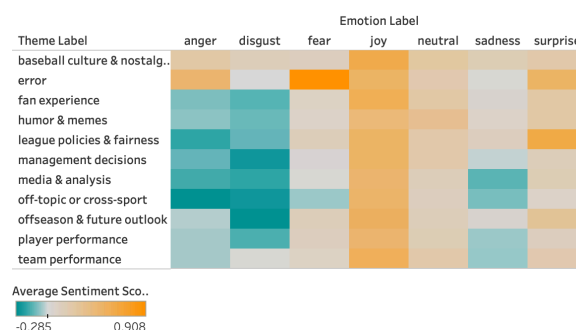
4.2.3 Discussion Themes and Their Sentiment Profiles

To assess the valence of each topic, the relationship between average sentiment and theme was explored.

Figure 6*Valence of Themes*

Positive valence was associated with themes such as “baseball culture & nostalgia”, “humor & memes”, and “fan experience” while neutral and negative valence was associated with themes such as “management decisions”, “media & analysis”, and “off-topic or cross-sport”. Interestingly, the lowest-valence category overall was “off-topic or cross-sport.”

As seen below, when emotions are crossed with themes, the lowest-scoring combinations were disgust in team-performance posts and sadness or fear within media/analysis discussions, both of which align with worse team performance. On the other hand, the highest-scoring combination was joy within culture/nostalgia and humor/memes, which correlate with more positive sentiment and better team performance.

Figure 7*Average Sentiment per Theme x Emotion*

4.2.4 Fanbase Volatility and Expectation Effects

Sentiment cannot capture the complex dynamics of fan discourse alone, the volatility was investigated (or standard deviation) of sentiment and residual sentiment after controlling for win percentage and run differential. This allowed notice to significant differences in fanbase “personality” that was not dependent on team performance.

Four broad fanbase profiles were observed:

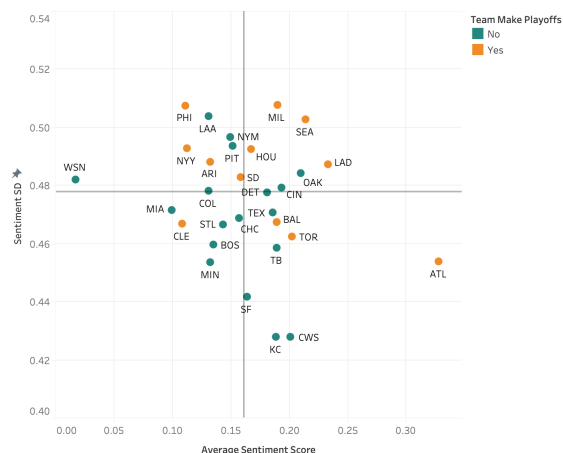
- **“Joyful diehards”** in the upper-right quadrant exhibited high sentiment and high volatility. Examples included the fanbases for the Milwaukee Brewers and the Seattle Mariners.
- **“Steady optimists”** in the lower-right quadrant, such as the fanbases for the

Atlanta Braves and the Toronto Blue Jays, showed high sentiment and low volatility.

- **“Emotional pessimists”** in the upper-left quadrant, exhibited low sentiment and high volatility and were found in the fanbases of teams such as the New York Yankees and Philadelphia Phillies.
- **“Stoic/disengaged”** fans in the lower-left quadrant, such as those of the Minnesota Twins and Boston Red Sox, reflected both low sentiment and low volatility.

Figure 8

Volatility of Sentiment vs. Average Sentiment



When values were adjusted for team performance, it was found that some fanbases

showed sentiment that was more positive than their performance predicted (ex. Braves, White Sox, Athletics), while others were systematically more negative (ex. Nationals, Yankees, Phillies). These findings highlight the influence of locational pressures, cultural expectations, and historical stories on fan sentiment and suggest that the bigger picture involves the interaction between those factors and current team performance.

5. Conclusions

The analysis revealed several clear patterns about what drives both on-field success and fan discourse in Major League Baseball. First, team performance models demonstrated that pitching and defensive metrics (especially runs allowed) were the strongest predictors of winning outcomes, substantially outweighing offensive statistics. Both the linear regression model for win percentage and the logistic regression model for playoff qualification achieved high accuracy, reinforcing the conclusion that preventing runs is the most consistent determinant of team success. In contrast, metrics such as payroll and certain pitching indicators like K/9 showed

counterintuitive effects once contextualized within multivariate models.

Second, the sentiment and emotion analyses indicated that fan attitudes reliably reflect real-world performance, though with weaker predictive power than statistical measures. Teams with winning records and playoff appearances exhibited more positive sentiment, while underperforming teams generated more negative sentiment. Emotion-specific patterns provided even stronger differentiation: joy was concentrated around successful teams, whereas disgust and fear were most common among struggling teams.

Finally, the exploration of fanbase behavior revealed meaningful variation in sentiment volatility and expectation-driven responses, identifying distinct fan personality clusters that persist even after controlling for performance. These findings suggest that while performance metrics explain outcomes on the field, emotional and thematic discourse adds important context for understanding how fans interpret and react to those outcomes.

Despite this, the study also had several notable limitations:

- **Sentiment analysis methods** are known to struggle with picking up sarcasm and irony. This limitation affected the emotion valence model's accuracy. As sentiment analysis models improve over time, the accuracy of this analysis can too.
- **The RAG-based chatbot** was deployed on CPU-only resources and optimized for lightweight experimentation, which limited the size of the vector database, volume of data that could be indexed, and complexity of the language model used.

This project opens the door to many possible future analyses:

- **Comparative league or level analysis:** Another direction could involve comparing major league data with minor league or collegiate baseball to see whether the same performance indicators predict winning at different levels of play.

- **Future longitudinal analyses:** Baseball as a sport has changed significantly over the years, which means that older data may not be directly comparable to more recent seasons. Future studies should therefore focus on updating analyses with newer datasets and examining how the relationships between statistics and winning outcomes change over time. This would help determine whether certain performance metrics remain consistent indicators of success or if new trends are emerging in the modern era of baseball. These analyses could also be used to understand how certain fan cultures, that were observed in this study, were formed and how they change over time.
- **Future Improvements for the Chatbot System:** While the current RAG-based chatbot successfully provides data-driven responses to baseball-related queries, several enhancements could improve its performance and scalability. With expanded infrastructure, the chatbot has potential to evolve into a

more powerful, comprehensive baseball analytics assistant. GPU-backed inference would allow for faster LLM generation, more efficient document embedding and retrieval, and integration of larger language models than the prototype system's current llama3:8b model. Larger models would improve the quality and accuracy of the chatbot's analytical explanations. Additionally, a logical next step would be to expand the knowledge base with historical data to support complex temporal and predictive analytics.

6. Author's Bios

Alisa Crowe earned her B.S. in Mathematics with a concentration in Data Science from UC Irvine in Fall 2023. After her graduation, she completed an internship in the AI Lab of a cybersecurity company, where she gained hands-on experience in applied machine learning. She is currently pursuing her M.S. in Big Data Analytics at San Diego State University, where she works as a Research Assistant and Lab Coordinator for the [AI4Business Lab](#). Her research focuses on

developing efficient computer vision models for pavement distress classification. Please feel free to connect with her on [LinkedIn](#) or via email at acrowe3069@sdsu.edu.

Rita Herfi received her bachelors degree in Mathematics with Emphasis in Computational Science in Spring of 2022 from San Diego State University. She currently is working full time at a healthcare administration company as an Accounting Associate and attending San Diego State University as a full time graduate student pursuing a masters in Big Data Analytics. As one of the participants of the [CHIARA](#) project from the Metabolism of Cities Living Lab (MOC-LLAB), she was able to contribute to creating a [dashboard](#) and writing two papers that were presented at international conferences. Rita can be contacted on [LinkedIn](#) or by email at rherfi2585@sdsu.edu

Samantha Keppler graduated from UC Santa Barbara in 2022 with a B.S. in Psychological and Brain Sciences. From there, she went on to work at a jury consulting firm, which ignited her passion for data processing and visualization. She is now pursuing her Masters in Big Data Analytics at San Diego

State University, where she works as a Research Assistant for the school's [Brain Development Imaging Laboratories](#) and a Graduate Assistant for the Big Data Analytics department.

Following her graduation in May 2026, she will assume a role as a Business Intelligence Analyst at Realty Income. Please feel free to connect with her on [LinkedIn](#) or contact her by email at skepler2948@sdsu.edu.

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8. Appendix

For more information, please click [here](#) to view the GitHub repository for this project.